

Project File Overview

This document summarizes the files in the current workspace. The project is split into two independent applications:

- `stm_creator-frontend` : React + Vite frontend for authentication, the STM graph editor, collaboration UI, model management, and the admin dashboard.
- `tern_backend` : TypeScript + Express + PostgreSQL + Socket.IO backend for REST APIs, authentication, permissions, migrations, collaboration, and tests.

Generated directories such as `node_modules/`, `dist/`, `.git/`, coverage output, logs, and PDF build profiles are not expanded file by file. The IDE tab `stm_creator-frontend/FRONTEND_LINT_DEBT.md` was not present on disk when this overview was prepared.

Production deployment:

- Frontend editor: `https://stm-8nizc.ondigitalocean.app/editor`
- Backend API used by the frontend source: `https://hammerhead-app-t8l9y.ondigitalocean.app`

Repository Root

File	Purpose
<code>package-lock.json</code>	Root lockfile with no real package dependencies.
<code>developer.pdf</code>	Existing generated developer PDF.
<code>DEVELOPER_FEATURE_RUNBOOK_EN.md.pdf</code>	Existing generated English developer runbook PDF.
<code>USER_FEATURE_GUIDE.md.pdf</code>	Existing generated user feature guide PDF.
<code>USER_FEATURE_GUIDE_EN.md.pdf</code>	Existing generated English user feature guide PDF.
<code>PROJECT_FILE_OVERVIEW_ZH.md</code>	Chinese project file overview.
<code>OPERATION_MANUAL_ZH.md</code>	Chinese operation manual.
<code>DEVELOPER_MANUAL_ZH.md</code>	Chinese developer manual.
<code>API_MANUAL_ZH.md</code>	Chinese API manual.

Frontend Project

Configuration and Documentation

File	Purpose
stm_creator-frontend/.env	Local Vite environment variables such as <code>VITE_API_BASE_URL</code> ; do not commit secrets.
stm_creator-frontend/.env.development	Development Vite environment variables.
stm_creator-frontend/.env.production	Production Vite environment variables.
stm_creator-frontend/.eslintrc.cjs	Frontend ESLint configuration for TypeScript, React Hooks, and React Refresh.
stm_creator-frontend/.gitignore	Ignores dependencies, build output, logs, and editor files.
stm_creator-frontend/.github/dependabot.yml	Dependabot configuration, currently focused on npm and <code>@xyflow/react</code> .
stm_creator-frontend/README.md	Original frontend README describing the React Flow STM Creator prototype and Vite workflow.
stm_creator-frontend/develop_Journal.md	Development journal for frontend refactors and features such as versioning, EKS import/export, comments, help modal, and PNG export.
stm_creator-frontend/template-first-workflow.md	Design note for the template-first workflow and matrix template layout.
stm_creator-frontend/LICENSE	MIT license.
stm_creator-frontend/package.json	Frontend dependencies and npm scripts.
stm_creator-frontend/package-lock.json	Frontend npm lockfile.
stm_creator-frontend/Procfile	Deployment startup command: <code>web: npm run start</code> .
stm_creator-frontend/index.html	Vite HTML entry.
stm_creator-frontend/tsconfig.json	Strict frontend TypeScript configuration.
stm_creator-frontend/tsconfig.node.json	TypeScript configuration for Vite config files.
stm_creator-frontend/vite.config.ts	Vite configuration with React plugin and sourcemaps.

Static Assets and Scripts

File	Purpose
stm_creator-frontend/public/anu-logo.svg	ANU logo asset.
stm_creator-frontend/public/BMRG_Rainforests.json	Sample STM JSON data used for demos and imports.
stm_creator-frontend/public/favicon.ico	Browser favicon.
stm_creator-frontend/public/tern.png	TERN image used by the admin UI.
stm_creator-frontend/scripts/connect-demo.mjs	Local connectivity demo for auth, model loading, and optional model saving.

Frontend Entry Points and Global Styles

File	Purpose
stm_creator-frontend/src/main.tsx	React root entry; wraps the app in <code>ErrorBoundary</code> and <code>ReactFlowProvider</code> .
stm_creator-frontend/src/App.tsx	Main route tree and graph editor composition layer. Wires the toolbar, React Flow canvas, comments, collaboration, node/edge modals, milestones, and admin route.
stm_creator-frontend/src/App.css	Main app, toolbar, editor, collaboration, and layout styles.
stm_creator-frontend/src/index.css	Global base styles.
stm_creator-frontend/src/EdgeStyles.css	React Flow transition edge styles.
stm_creator-frontend/src/SwimlaneStyle.css	Swimlane-related styling.
stm_creator-frontend/src/SwimLanes.tsx	Swimlane/experimental layout component.
stm_creator-frontend/src/vite-env.d.ts	Vite TypeScript declarations.

API Client and Authentication

File	Purpose
stm_creator-frontend/src/app/auth/api.ts	API base URL selection, auth calls, JWT storage, and shared <code>apiFetch</code> . Defaults to the DigitalOcean backend unless overridden.
stm_creator-frontend/src/app/auth/AuthPage.tsx	Login/signup UI, guest entry, and model selection after login.
stm_creator-frontend/src/app/auth/EcosystemPanel.tsx	Visual ecosystem panel shown on the auth page.
stm_creator-frontend/src/app/api/comments.ts	REST client for comments.
stm_creator-frontend/src/app/api/drivers.ts	REST client for driver autocomplete.
stm_creator-frontend/src/app/api/locks.ts	REST client for model-level locks.
stm_creator-frontend/src/app/api/models.ts	REST client for models, templates, permissions, and milestones.

Pages

File	Purpose
stm_creator-frontend/src/pages/Home.tsx	Landing/home page introducing the product and routing users into the editor.
stm_creator-frontend/src/pages/Home.css	Home page styles.
stm_creator-frontend/src/pages/AdminDashboard.tsx	Admin dashboard for stats, users, sessions, and audit logs.
stm_creator-frontend/src/pages/AdminDashboard.module.css	CSS Module for the admin page.
stm_creator-frontend/src/pages/VerifyEmail.tsx	Email verification page that consumes a URL token.
stm_creator-frontend/src/pages/Forbidden.tsx	Forbidden-access page.
stm_creator-frontend/src/pages/NotFound.tsx	Not-found page.
stm_creator-frontend/src/pages/NotFound.css	Not-found page styles.

Shared Components

File	Purpose
stm_creator-frontend/src/components/ErrorBoundary.tsx	React error boundary.
stm_creator-frontend/src/components/CreateModelModal.tsx	Modal for creating a blank model or cloning a template.
stm_creator-frontend/src/components/ModelPermissionsModal.tsx	Model sharing and permission-management modal.
stm_creator-frontend/src/components/toolbar/ToolbarDropdown.tsx	Reusable toolbar dropdown.
stm_creator-frontend/src/components/admin/ConfirmModal.tsx	Admin confirmation modal.
stm_creator-frontend/src/components/admin/ProtectedAdminRoute.tsx	Client-side admin route guard.
stm_creator-frontend/src/components/admin/StatsCard.tsx	Admin statistics card.
stm_creator-frontend/src/components/admin/UsersTable.tsx	Admin user table.

Editor Components

File	Purpose
stm_creator-frontend/src/app/components/GraphToolbar.tsx	Main editor toolbar with Canvas, Model, File, View, lock, admin, share, models, help, comments, and logout actions.
stm_creator-frontend/src/app/components/CanvasContextMenu.tsx	Right-click menu for editing/deleting states and transitions.
stm_creator-frontend/src/app/components/CommentPanel.tsx	Right-side comment panel with mentions, resolve/delete actions, and pagination.
stm_creator-frontend/src/app/components/EdgeCreationHint.tsx	Hint shown while edge creation mode is active.
stm_creator-frontend/src/app/components/ErrorMessage.tsx	Editor error state.
stm_creator-frontend/src/app/components/HelpModal.tsx	Help/support modal that generates an email draft.
stm_creator-frontend/src/app/components/LoadingState.tsx	Editor loading state.
stm_creator-frontend/src/app/components/MilestoneModal.tsx	Unified milestone modal for save, restore, and delete actions.
stm_creator-frontend/src/app/components/ModelListModal.tsx	Model browser modal.
stm_creator-frontend/src/app/components/ModelSelectionModal.tsx	Post-login model selection modal.
stm_creator-frontend/src/app/components/TipsPanel.tsx	Tips panel.
stm_creator-frontend/src/app/components/VersionComparisonModal.tsx	Local version comparison modal.
stm_creator-frontend/src/app/components/VersionComparisonModal.css	Version comparison styles.
stm_creator-frontend/src/app/components/VersionManagerModal.tsx	Older local version manager component, mostly superseded by <code>MilestoneModal</code> .
stm_creator-frontend/src/app/types.ts	Shared app-layer types.

Editor Hooks and Graph Logic

File	Purpose
stm_creator-frontend/src/app/hooks/useGraphEditor.ts	Main graph-editor orchestration hook.
stm_creator-frontend/src/app/hooks/useGraphEditor.types.ts	Hook input/output types.
stm_creator-frontend/src/app/hooks/useGraphBaseState.ts	Base graph editor state.
stm_creator-frontend/src/app/hooks/graphConstants.ts	React Flow node/edge types and default edge options.
stm_creator-frontend/src/app/hooks/graphNodes.ts	Node creation, editing, duplication, and click handling.
stm_creator-frontend/src/app/hooks/graphTransitions.ts	Transition creation and transition handler factories.
stm_creator-frontend/src/app/hooks/edgeHandlers.ts	Edge click/double-click, edge changes, and transition save handling.
stm_creator-frontend/src/app/hooks/graphMutations.ts	Local and remote graph patch helpers.
stm_creator-frontend/src/app/hooks/graphRebuilder.ts	Rebuilds React Flow edges from model transitions.
stm_creator-frontend/src/app/hooks/graphFilters.ts	Transition filtering actions.
stm_creator-frontend/src/app/hooks/graphModel.ts	Model load/save/refresh and relay layout entry points.
stm_creator-frontend/src/app/hooks/graphDeletes.ts	Model, state, and transition deletion flows.
stm_creator-frontend/src/app/hooks/graphImportExport.ts	EKS JSON import/export logic.
stm_creator-frontend/src/app/hooks/graphVersions.ts	Local snapshot versioning logic.
stm_creator-frontend/src/app/hooks/graphLayout.ts	Graph layout strategies.
stm_creator-frontend/src/app/hooks/graph-utils.ts	Small graph utilities such as ID parsing.

Collaboration, Roles, and Permissions

File	Purpose
stm_creator-frontend/src/collab/socket.ts	Socket.IO client wrapper for presence, cursors, locks, and field patches.
stm_creator-frontend/src/constants/roles.ts	Frontend global and model role constants.
stm_creator-frontend/src/utils/permissions.ts	Client-side permission helpers.

Nodes, Edges, and Transitions

File	Purpose
stm_creator-frontend/src/nodes/customNode.tsx	Custom state node renderer.
stm_creator-frontend/src/nodes/customNode.css	State node styles.
stm_creator-frontend/src/nodes/nodeModal.tsx	State create/edit modal with templates, VAST class, condition range, notes, and image upload.
stm_creator-frontend/src/nodes/nodeModal.css	Node modal styles.
stm_creator-frontend/src/nodes/templates.ts	Frontend state template catalog and template-reference helpers.
stm_creator-frontend/src/nodes/types.ts	Node type definitions.
stm_creator-frontend/src/nodes/PositionLoggerNode.tsx	Position debugging/logging node.
stm_creator-frontend/src/nodes/index.ts	Node barrel export.
stm_creator-frontend/src/edges/customEdge.tsx	Custom transition edge renderer.
stm_creator-frontend/src/edges/customEdge.css	Transition edge styles.
stm_creator-frontend/src/edges/index.ts	Edge barrel export.
stm_creator-frontend/src/transitions/transitionModal.tsx	Transition edit modal with basic fields, delta calculation, causal-chain editing, driver search, and preconditions.
stm_creator-frontend/src/transitions/transitionModal.css	Transition modal styles.

Extensions and Onboarding

File	Purpose
stm_creator-frontend/src/extensions/TransitionFilterPanel.tsx	Transition filter panel.
stm_creator-frontend/src/extensions/extensions.css	Shared extension styles.
stm_creator-frontend/src/extensions/index.ts	Extension barrel export.
stm_creator-frontend/src/extensions/i18n/index.ts	i18n utility entry.
stm_creator-frontend/src/extensions/i18n/fmt.ts	Formatting helpers.
stm_creator-frontend/src/extensions/i18n/uiStrings.ts	UI string constants.
stm_creator-frontend/src/extensions/onboarding/Tour.tsx	Guided tour component.
stm_creator-frontend/src/extensions/onboarding/coachmarks.ts	Guided tour steps.
stm_creator-frontend/src/extensions/onboarding/useOnboarding.ts	Onboarding state hook.
stm_creator-frontend/src/extensions/onboarding/Checklist.tsx	Onboarding checklist.
stm_creator-frontend/src/extensions/onboarding/PresetsPanel.tsx	Presets panel.
stm_creator-frontend/src/extensions/onboarding/index.ts	Onboarding barrel export.

Data Conversion and Utilities

File	Purpose
stm_creator-frontend/src/utils/dataLoader.tsx	Resolves the current model name, loads models, saves models, and prepares save payloads.
stm_creator-frontend/src/utils/eksJson.ts	Converts between <code>BMRGData</code> and EKS JSON.
stm_creator-frontend/src/utils/layoutStrategies.ts	Layout strategy definitions.
stm_creator-frontend/src/utils/conditionColours.ts	VAST condition-class color definitions.
stm_creator-frontend/src/utils/versionStorage.ts	Local snapshot storage in <code>localStorage</code> .
stm_creator-frontend/src/utils/stateTransition/index.ts	State-transition utility barrel export.
stm_creator-frontend/src/utils/stateTransition/types.ts	Frontend <code>BMRGData</code> , <code>StateData</code> , and <code>TransitionData</code> types.
stm_creator-frontend/src/utils/stateTransition/helpers.ts	Shared helpers including transition delta calculation.
stm_creator-frontend/src/utils/stateTransition/nodes.ts	Converts state data into React Flow nodes.
stm_creator-frontend/src/utils/stateTransition/edges.ts	Converts transition data into React Flow edges.
stm_creator-frontend/src/utils/stateTransition/layout.ts	State node layout and geometry helpers.

Backend Project

Configuration, Documentation, and Deployment

File	Purpose
tern_backend/.dockerignore	Docker build ignore rules.
tern_backend/.env.example	Backend environment variable template.
tern_backend/.gitignore	Backend Git ignore rules.
tern_backend/.github/workflows/ci.yml	Backend CI: lint, type check, build, and Jest tests.
tern_backend/README.md	Backend overview, features, stack, structure, and startup instructions.
tern_backend/E2E_TEST_REPORT.md	E2E/API test report.
tern_backend/docs/API.md	Existing English API reference. Some parts are older than current source.
tern_backend/docs/WEBSOCKET.md	Existing English Socket.IO documentation.
tern_backend/docs/MIGRATIONS.md	Existing migration runbook.
tern_backend/package.json	Backend dependencies and scripts.
tern_backend/package-lock.json	Backend npm lockfile.
tern_backend/tsconfig.json	Backend TypeScript configuration.
tern_backend/eslint.config.mts	Backend ESLint flat config.
tern_backend/jest.config.ts	Jest configuration.
tern_backend/jest.setup.ts	Jest environment defaults.
tern_backend/database.json	node-pg-migrate database configuration.
tern_backend/Dockerfile	Multi-stage Dockerfile for build, migrate, dev, and runtime.
tern_backend/docker-compose.yml	Local PostGIS, app, and migration services.

Postman

File	Purpose
tern_backend/postman/TERN_API.collection.json	Postman/Newman API test collection.
tern_backend/postman/local.environment.json	Local Newman environment.
tern_backend/postman/production.environment.json	Production Newman environment.

Database Migrations

File	Purpose
1758438322642_tern-postgresql-migration.mjs	Initial PostgreSQL/PostGIS schema.
1758600000000_collab-tables.mjs	Collaboration tables: activity, locks, comments, milestones, mentions.
1758700000000_fix-transition-id-unique.mjs	Transition business ID uniqueness per model.
1758800000000_model-permissions.mjs	Model-level permissions and <code>model_role</code> .
1758800000000_state-node-positions.mjs	State node coordinates.
1758850000000_email-verification.mjs	Email verification fields and token table.
1758900000000_add-is-template.mjs	Template flag on <code>stmmodel</code> .
1758900000000_admin-tables.mjs	Sessions and audit log tables.
1759000000000_add-review-lock-columns.mjs	Review-lock columns on <code>stmmodel</code> .
1759100000000_seed-default-templates.mjs	Default template seed data.
1759200000000_rename-default-templates.mjs	Renames defaults to <code>Woodlands</code> , <code>Grasslands</code> , <code>Shrublands</code> .
1759300000000_add-owner-model-role.mjs	Adds <code>owner</code> to <code>model_role</code> .
1759400000000_add-lock-type.mjs	Adds <code>lock_type</code> to collaboration locks.
1759500000000_cascade-and-indexes.mjs	Permission foreign keys, indexes, and cascade behavior.

Backend Entry Points and Configuration

File	Purpose
<code>tern_backend/src/index.ts</code>	HTTP server entry; initializes Express, Socket.IO, socket auth, collaboration handlers, and lock cleanup.
<code>tern_backend/src/app.ts</code>	Express middleware, CORS, Swagger, route mounting, error handling, and 404 redirect.
<code>tern_backend/src/socket.ts</code>	Socket.IO initialization and global lazy proxy.
<code>tern_backend/src/config/database.ts</code>	PostgreSQL Pool setup.
<code>tern_backend/src/config/env.ts</code>	Required environment checks and frontend/backend URL configuration.
<code>tern_backend/src/constants/roles.ts</code>	Global and model role constants.
<code>tern_backend/src/errors.ts</code>	Application error classes and codes.
<code>tern_backend/src/swagger/swagger.ts</code>	OpenAPI/Swagger generation.

Middleware and Validation

File	Purpose
<code>auth.middleware.ts</code>	JWT bearer authentication and verified-email check.
<code>role.middleware.ts</code>	Global role guard.
<code>model-permission.middleware.ts</code>	Model-level role guard and owner/admin permission guard.
<code>requireAdmin.ts</code>	Admin-only guard.
<code>rateLimit.ts</code>	In-memory rate limiters.
<code>requestId.middleware.ts</code>	Request ID assignment.
<code>error.middleware.ts</code>	Unified error envelope.
<code>validation/validate.ts</code>	Zod validation middleware.
<code>validation/auth.schemas.ts</code>	Auth request schemas.
<code>validation/models.validation.ts</code>	Model route schemas.
<code>validation/drivers.validation.ts</code>	Driver route schemas.

REST Routes

File	Purpose
<code>routes/auth.ts</code>	<code>/auth</code> : signup, verify, resend verification, login.
<code>routes/models.ts</code>	<code>/models</code> : model lifecycle, templates, save/delete, review locks.
<code>routes/permissions.ts</code>	<code>/models/:name/permissions</code> : model-level permissions.
<code>routes/locks.ts</code>	<code>/models/:name/lock*</code> : REST model locks.
<code>routes/drivers.ts</code>	<code>/drivers</code> : driver search and sub-drivers.
<code>routes/collab.ts</code>	<code>/collab</code> : activity, comments, milestones.
<code>routes/notifications.ts</code>	<code>/notifications</code> : mention notifications.
<code>routes/admin.ts</code>	<code>/api/admin</code> : admin stats, users, sessions, audit logs.

Services

File	Purpose
<code>services/auth.service.ts</code>	User creation, login, verification token creation/consumption, contributor linking.
<code>services/email.service.ts</code>	Gmail/Nodemailer verification emails.
<code>services/users.service.ts</code>	User service helper/placeholder.
<code>services/models/show.service.ts</code>	Model listing and full model aggregation.
<code>services/models/save.service.ts</code>	Model save/update, state/transition/driver upserts, template cloning, owner grants.
<code>services/models/remove.service.ts</code>	Model, state, and transition deletion.
<code>services/models/reviewLock.service.ts</code>	Review-lock checks and mutations.
<code>services/permissions/model-permissions.service.ts</code>	Model permission list/grant/revoke/lookups.
<code>services/drivers/search.service.ts</code>	Driver and sub-driver queries.
<code>services/collab/activity.service.ts</code>	Activity logging and retrieval.
<code>services/collab/comments.service.ts</code>	Comment create/list/resolve/delete.
<code>services/collab/locks.service.ts</code>	Collaboration lock service.
<code>services/collab/mentions.service.ts</code>	Mention notifications.
<code>services/collab/milestones.service.ts</code>	Milestone create/list/get/delete.

Socket.IO Collaboration

File	Purpose
collab/auth.middleware.ts	Socket.IO handshake JWT authentication.
collab/socket.ts	Collaboration event handlers.
collab/roomManager.ts	Room membership, online users, and activity broadcasting.
collab/lockCache.ts	In-memory entity lock cache.
collab/lockCleanup.ts	Periodic expired-lock cleanup.
collab/types.ts	Collaboration types.

Types and Utilities

File	Purpose
types/auth.types.ts	Auth DTO and user types.
types/models.types.ts	Backend model types.
types/drivers.types.ts	Driver result types.
types/permissions.types.ts	Model permission types.
types/express.d.ts	Express type extensions.
utils/hash.ts	bcrypt + pepper hashing.
utils/jwt.ts	JWT signing and verification.
utils/transition.utils.ts	Transition delta calculation.
utils/auditLog.ts	Admin audit log helper.

Tests

Area	Files
Integration tests	tests/integration/*.test.ts for admin, auth, collab, CORS, drivers, locks, models, notifications, permissions, and Socket.IO.
Unit tests	tests/unit/**/*.test.ts for middleware, services, collaboration helpers, JWT, hashing, and transition utilities.

Generated Backend Directories

Directory	Purpose
tern_backend/.git/	Git repository metadata.
tern_backend/node_modules/	Installed dependencies.
tern_backend/dist/	TypeScript build output.
tern_backend/coverage/	Jest coverage output.
tern_backend/results/	Newman output.